# THE INJECTIVE OBJECTS OF ADEQUATE FUNCTORS ARE THE PURE INJECTIVE MODULES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- The category of adequate functors has enough injectives. Same chapter, Lemma lemma-enough-injectives, invoked at proof lines 2977-2979. Provides the injective object used in the converse direction.
- Universally exact sequences of modules give exact sequences of the associated module-valued functors, and purity means splitting. The characterisation of pure injective modules used in the argument. Identifies the module I(A) as pure injective in one direction and produces the splitting in the other.

*Source reference.* Original lemma 06Z7: sources/stacks-adequate/pinned-original-source.tex, source lines 1086–1086. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $A$ be a ring. Let $\mathcal{A}$ be the category of adequate functors on $Alg_A$. The injective objects of $\mathcal{A}$ are exactly the functors $\underline{I}$ where $I$ is a pure injective $A$-module.*

*Proof.* Let $I$ be an injective object of $\mathcal{A}$. Choose an embedding $I \to \underline{M}$ for some $A$-module $M$. As $I$ is injective we see that $\underline{M} = I \oplus F$ for some module-valued functor $F$. Then $M = I(A) \oplus F(A)$ and it follows that $I = \underline{I(A)}$. Thus we see that any injective object is of the form $\underline{I}$ for some $A$-module $I$. It is clear that the module $I$ has to be pure injective since any universally exact sequence $0 \to M \to N \to L \to 0$ gives rise to an exact sequence $0 \to \underline{M} \to \underline{N} \to \underline{L} \to 0$ of $\mathcal{A}$.

Finally, suppose that $I$ is a pure injective $A$-module. Choose an embedding $\underline{I} \to J$ into an injective object of $\mathcal{A}$ (see Lemma 06Z7). We have seen above that $J = \underline{I'}$ for some $A$-module $I'$ which is pure injective. As $\underline{I} \to \underline{I'}$ is injective the map $I \to I'$ is universally injective. By assumption on $I$ it splits. Hence $\underline{I}$ is a summand of $J = \underline{I'}$ whence an injective object of the category $\mathcal{A}$. □

*Date*: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.